

1.

A random sample of 16 emergency reports was selected from the files of an ambulance service. The mean time (computed from the sample data) required for ambulances to reach their destinations was 13 minutes. Assume that the population of times is normally distributed with a variance of 9. Can we conclude at the .05 level of significance that the population mean is greater than 10 minutes?

Solution:

Here, $n = 16$, $\bar{x} = 13$, $\sigma^2 = 9$, $\alpha = 0.05$

Hypotheses:

$$H_0: \mu = 10$$

$$H_a: \mu > 10$$

Critical value: $z_{1-\alpha} = z_{0.95} = 1.64$

Test Statistic:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{13 - 10}{3/\sqrt{16}} = 4$$

Decision: Since $z_{cal} = 4 > z_{0.95} = 1.64$, we reject the null hypothesis at 5% level of significance and conclude that the mean time in the population is greater than 10 minutes.

2.

Suppose it is known that the IQ scores of a certain population of adults are approximately normally distributed with a standard deviation of 15. A simple random sample of 25 adults drawn from this population had a mean IQ score of 105. On the basis of these data can we conclude that the mean IQ score for the population is not 100? Let the probability of committing a type I error be .05.

Solution:

Here, $n = 25$, $\bar{x} = 105$, $\sigma = 15$, $\alpha = 0.05$

Hypotheses:

$$H_0: \mu = 100$$

$$H_a: \mu \neq 100$$

Critical value: $z_{1-\frac{\alpha}{2}} = z_{0.975} = 1.96$ or $z_{\frac{\alpha}{2}} = z_{0.05} = -1.96$

Test Statistic:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{105 - 100}{15/\sqrt{25}} = 1.67$$

Decision: Since $z_{cal} = 1.67 \not> z_{0.975} = 1.96$, we failed to reject the null hypothesis at 5% level of significance and conclude that the mean IQ score in the population is not 100.